

JIGAR ROHIT

MCP INTEGRATION DEVELOPER · PYTHON BACKEND ENGINEER

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PROFESSIONAL SUMMARY

Results-driven Python Backend Engineer and MCP Integration Developer specializing in high-volume data engineering pipelines, enterprise REST APIs, and multi-model AI system orchestration. Proven track record architecting production ETL batch scripts handling ~2,000,000 records, building secure FastAPI backends with PostgreSQL and Flyway schema versioning, and deploying Model Context Protocol (MCP) architectures with strict credential sandboxing and sub-second routing heuristics.

TECHNICAL PROFICIENCIES

Languages & Frameworks:	Python 3.11, FastAPI, SQL, TypeScript, React 19, Pydantic v2, Vite
Databases & Migrations:	PostgreSQL, Flyway Migrations, SQLAlchemy Core, Schema Normalization, Relational Indexing
AI / LLM & Protocols:	Model Context Protocol (MCP), JSON-RPC 2.0, LiteLLM, Groq, Anthropic, Gemini, Dynamic Model Routing
Data Engineering:	High-Volume ETL (~2M Records), Chunked Batch Processing, Deduplication, Anomaly Quarantine, Data Cleansing
Architecture & DevOps:	RBAC Authorization, Asynchronous I/O (asyncio), Docker, CI/CD GitHub Actions, Uvicorn, REST API Design

FEATURED ENGINEERING PROJECTS

Alef Migration — High-Scale Data Engineering Pipeline (~2,000,000 Records)

Python 3.11 · Batch Ingestion · ETL · Data Normalization · Anomaly Quarantine

- Engineered advanced Python automation batch scripts to prepare and migrate ~2,000,000 records between disparate database systems with divergent schema acceptance criteria.
- Built a modular 8-stage transformation pipeline incorporating source ingestion, unicode normalization, date reformatting, business constraint validation, and cross-reference deduplication.
- Implemented automated anomaly quarantine logic to isolate malformed records into dedicated audit ledgers without interrupting active batch execution, maintaining 100% pipeline fault tolerance.
- Designed memory-safe generator streams with atomic chunked batch commits, completely eliminating Out-Of-Memory (OOM) failures under extreme data loads.

IMS: Inventory Management System & IT Ticketing Platform

FastAPI · PostgreSQL · Flyway Migrations · React 19 · RBAC Authorization

- Architected and deployed a multi-tenant enterprise hardware asset tracking and IT ticket resolution platform with FastAPI and PostgreSQL.
- Integrated Flyway database version control to execute reproducible, deterministic DDL schema migrations across deployment staging tiers.
- Enforced strict Role-Based Access Control (RBAC) across Admin, Support Engineer, and Developer roles, restricting schema migration and hardware allocation to authorized personas.
- Engineered Pydantic v2 validation middleware with strict input sanitization, parameterized SQL queries, and deterministic operational audit logs.

NamoGPT — Multi-Model AI Orchestrator & Gateway

LiteLLM · Multi-Provider Routing · Model Context Protocol (MCP) · Client Credential Isolation

- Developed a multi-provider AI gateway unifying 11+ model providers and inference engines (Anthropic Claude, Groq LPU, Google Gemini, NVIDIA NIM, Ollama) behind a unified interface.
- Implemented dynamic heuristic routing analyzing prompt intent (coding AST accuracy, low-latency sub-150ms TTFT, extended 2M-token document context, and multi-step reasoning).
- Constructed client-isolated API key security architecture ensuring zero server-side credential persistence or sensitive environment leaks.
- Engineered resilient multi-tier fallback chains automatically redirecting interrupted requests to alternate providers without client session termination.

ENGINEERING PRINCIPLES & GOVERNANCE

Architecture Tenets: Reliability Before Complexity · Explicit Input Validation & Sanitization · Clean Separation of Concerns · Zero-Trust Credential Security · Atomic Database Transactions · Comprehensive Automated Testing (Pytest unit suites & CI/CD pipeline integration).